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Proposal for a Master's Degree in Computer Science

Heart Disease Prediction and Detection

Using Artificial Intelligence Techniques

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Abstract

Cardiovascular disease remains the foremost cause of mortality worldwide, accounting for a substantial proportion of global deaths each year. A large share of these deaths is preventable through timely and accurate risk assessment, which makes the development of reliable predictive tools an urgent priority in clinical decision support. Contemporary AI-based approaches to heart disease prediction, however, frequently depend on a single model that may struggle to capture the complex, non-linear interactions characteristic of medical tabular data and that generalizes poorly across heterogeneous patient populations.

This proposal presents a research plan for the design, implementation, and rigorous evaluation of a hybrid AI framework for heart disease prediction. The proposed framework integrates two complementary AI models, each selected for its ability to capture distinct patterns within structured clinical data, and combines their outputs to deliver superior predictive performance and improved generalization relative to individual baseline models. The research will benchmark the proposed approach against established methods using standard clinical evaluation metrics, including accuracy, precision, recall, and the F1-score, on publicly available structured heart disease datasets.

Keywords: Heart Disease Prediction, Cardiovascular Risk Assessment, Artificial Intelligence, Hybrid AI Framework, Ensemble Learning, Tabular Medical Data, Clinical Decision Support.

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1. Introduction

Heart disease, encompassing the broad family of cardiovascular conditions affecting the heart and blood vessels, is one of the leading causes of death worldwide and a major public health concern. It affects millions of people each year and, when left undetected or poorly managed, can progress to severe and frequently fatal complications. Among the most consequential of these complications are heart attack, in which blood flow to a portion of the heart muscle is abruptly interrupted; stroke, in which the blood supply to the brain is compromised; and heart failure, in which the heart is progressively unable to pump blood effectively to meet the body's needs.

The clinical and societal burden imposed by these conditions underscores the importance of prevention and timely intervention. Early detection and proper management are essential to reduce risks, improve patients' quality of life, and increase survival rates. Advances in artificial intelligence and machine learning have created an opportunity to support clinicians in this task by extracting predictive insight from routinely collected patient data, thereby enabling earlier and more accurate identification of individuals at elevated cardiovascular risk.

2. Motivation

The motivation for this research is grounded in the scale of the cardiovascular disease burden and in the demonstrable potential of preventive intervention. The following statistics, reported by the World Health Organization and related epidemiological sources, illustrate the magnitude of the problem and the opportunity that early prediction presents:

- **Leading cause of death:** Cardiovascular disease is the single leading cause of death globally, exceeding the mortality attributable to any other category of disease.
- **Approximately 17.9 million deaths per year:** An estimated 17.9 million people die from cardiovascular disease each year worldwide, reflecting the pervasive and sustained nature of the threat.
- **32% of all global deaths:** Cardiovascular disease accounts for roughly thirty-two percent of all deaths recorded worldwide, a proportion that highlights its dominance among causes of mortality.
- **Up to 80% preventable:** A substantial share of premature cardiovascular deaths is considered preventable through early detection, risk-factor management, and timely intervention, establishing a compelling case for accurate predictive tools.

Taken together, these figures motivate the development of intelligent, data-driven predictive systems capable of supporting early diagnosis. Even modest improvements in predictive accuracy can

translate into meaningful reductions in preventable mortality when deployed at scale within clinical workflows.

Beyond the medical imperative, this research is equally motivated by identifiable limitations in the current technical landscape. A review of existing literature reveals that, while many prior studies report high predictive accuracy, a substantial number achieve this through approaches that sacrifice interpretability, generalize poorly beyond their training datasets, or exploit a single AI technique whose inductive biases may not suit the full complexity of clinical data. This research aims to advance the state of the art by designing a hybrid framework that combines the complementary strengths of multiple AI paradigms, thereby improving not only accuracy but also robustness and reliability across diverse patient populations.

3. Problem Statement

Despite considerable progress in applying machine learning to heart disease prediction, existing approaches exhibit several persistent limitations that constrain their predictive power and clinical reliability. These limitations define the problem that this research seeks to address.

3.1 Limited Diversity in Predictive Approaches

While individual AI models can achieve strong predictive performance, their effectiveness is inherently bounded by the inductive biases of a single underlying algorithm. In practice, no single approach captures the full spectrum of patterns present in complex clinical data. The absence of complementary model perspectives limits the capacity of existing systems to deliver consistently reliable predictions across the varied scenarios encountered in real-world clinical settings.

3.2 Difficulty Capturing Feature Interaction Complexity

Existing models frequently struggle to capture the complex, non-linear feature interactions that are inherent in medical tabular data. Physiological and clinical variables interact in intricate ways, and predictive performance depends critically on a model's ability to represent these higher-order relationships. Methods that fail to model such interactions adequately leave valuable diagnostic signal unexploited.

3.3 Limited Generalization Across Diverse Patient Data

Many models fail to generalize across diverse patient populations characterized by varying demographics and clinical profiles. A model that performs well on a particular dataset may degrade substantially when applied to populations that differ in age distribution, comorbidity profile, or data-collection protocol. This lack of generalization, often tied to the limited diversity of available training datasets, poses a significant barrier to safe and equitable clinical deployment.

4. Research Objectives

In response to the limitations identified above, this research pursues the following primary objectives:

1. **Integration of advanced algorithms.** To design a hybrid AI framework that integrates two complementary models, combining their respective strengths to achieve superior predictive performance on structured clinical data.
2. **Comparative performance evaluation.** To rigorously benchmark the proposed hybrid framework against state-of-the-art individual models across a consistent set of standard evaluation metrics, establishing the relative advantage of the hybrid approach under controlled experimental conditions.
3. **Complex clinical pattern learning.** To capture non-linear feature interactions and long-range dependencies within patient data, with the goal of improving diagnostic accuracy and enhancing the model's ability to represent clinically meaningful relationships.

5. Literature Review

A review of recent literature reveals an active and rapidly evolving body of work on machine learning for heart disease and cardiovascular risk prediction. The studies surveyed below span hybrid transformer architectures, ensemble learning strategies, and explainable artificial intelligence techniques. Collectively, they establish both the strong performance attainable on benchmark datasets and the recurring limitations that motivate the present research. The comparative analysis is summarized in Table 1, which contrasts each study with respect to its objective, methodology, dataset, reported results, and principal limitation.

Table 1. Comparative analysis of recent studies on machine learning for heart disease prediction.

Paper	Objective	Technique	Dataset	Results	Limitation
CardioTabNet: A Novel Hybrid Transformer Model for Heart Disease Prediction using Tabular Medical Data (2025)	Enhance cardiovascular disease predictive accuracy by leveraging a Tab Transformer to extract meaningful feature spaces from tabular clinical data.	Tab Transformer; Random Forest (feature ranking); Extra Tree Classifier optimized with Optuna.	IEEE Data Port Heart Disease Dataset.	Accuracy 94.08%; AUC 95.0%; Precision 92.84%; Recall 97.37%; F1-Score 94.47%.	Relatively modest dataset size, potentially limiting generalization to broader populations.
HeartEnsembleNet: An Innovative Hybrid Ensemble Learning Approach for Cardiovascular Risk Prediction (2025)	Identify risk factors contributing to cardiac dysfunction using a hybrid ensemble learning model.	Hybrid Random Forest Linear Model (HRFLM); stacking; voting; RF + KNN.	Kaggle Cardiovascular Disease Dataset.	Accuracy 92.95%; Precision 93.08%; Recall 92.88%; F1-Score 92.47%.	Lacks specific cardiovascular diagnostic markers (ECG, echocardiography, troponins, NT-proBNP).
Comparative Study of Transformer-Based Models for Cardiovascular Disease Risk Stratification with Tabular Biomarker Data (2026)	Compare traditional ML, deep learning, and transformer-based models using statistical and clinical biomarker data for CVD risk prediction.	FT-Transformer; SAINT; Tab Transformer; GRU; LSTM.	Cardiovascular Imaging Network at Queen's University cohort (n = 959 real patients; augmented to n = 3,060 via SMOTE; 21 biomarkers).	AUC up to 0.9728 (Tab Transformer) and 0.9489 (FT-Transformer); Accuracy up to 95.18% (Tab Transformer) and 89.18% (FT-Transformer).	High computational cost; reliance on synthetic sampling (SMOTE); lacks variables to compare against standard CVD calculators.
An Interpretable Ensemble Learning Framework (IELF) Balancing Predictive Accuracy with Clinical Interpretability (2025)	Develop an interpretable ensemble learning framework that balances predictive accuracy with clinical interpretability.	Explainable Boosting Machines (EBM); XGBoost; soft-voting ensemble; SHAP; LIME.	Cleveland (n = 303; 13 features); Framingham (n = 4,240; 15 features).	Cleveland: Accuracy 88.5%, AUC 96.0%. Framingham: Accuracy 82.6%, AUC 68.6%.	Datasets are dated and lack contemporary demographic diversity; harmonization obscures population-specific nuances.
Ensemble Learning with Explainable AI for Improved Heart Disease Prediction Based on Multiple Datasets (2025)	Enhance the robustness and accuracy of heart disease prediction by applying stacking and voting ensemble methods alongside Explainable AI.	Stacking (LR meta-model); voting; LR, ET, RF, CatBoost, XGBoost, LightGBM; SHAP.	HDDC (Kaggle / Cleveland-Hungary, n = 1,190; 12 attributes); UHDD (Kaggle, n = 1,025; 14 attributes).	Stacking Accuracy ~91% (HDDC) and ~98% (UHDD); ROC-AUC 0.92 (HDDC) and 0.97 (UHDD).	Generalizability constrained by public datasets; high computational complexity for tuning; some feature anomalies in SHAP analysis.
Harnessing Transformer Models for Cardiovascular Disease Prediction: A Comparison with Conventional Methods (2025)	Systematically evaluate transformer-based models against conventional ML models across datasets of varying scale and class imbalance.	FT-Transformer; SAINT; TabNet; Tab Transformer; XGBoost; LightGBM; RF; SVM; LR; NB; SMOTETomek; MICE.	UCI Heart Disease (n = 1,025); Framingham (n = 4,240); Kaggle (n = 70,000).	UCI: FT-Transformer (Acc 0.98, AUC 0.99), XGBoost (Acc 0.97, AUC 0.98). Kaggle: FT-Transformer (Acc 0.74, AUC 0.80), XGBoost (Acc 0.73, AUC 0.79).	Transformer models degrade significantly on severely imbalanced datasets compared with stable classical baselines.
Heart Disease Prediction Using a Hybrid TabNet Architecture with Stacked Ensemble Learning (2025)	Propose a stacked ensemble framework integrating a tabular deep learning model and robust tree-based methods to handle noisy patient data.	TabNet; XGBoost; stacked ensemble (Logistic Regression / SVM meta-learner).	Kaggle CVD (n = 70,000; 11 features); UCI CVD (n = 920; 11 features).	Kaggle: Accuracy 80.20%, F1 78.42%. UCI: Accuracy 95.20%, F1 91.92%.	Evaluated on benchmark datasets lacking real-world EHR noise; limited representation of diverse ethnicities and age groups.

Paper	Objective	Technique	Dataset	Results	Limitation
Optimized Ensemble Learning Approach with Explainable AI for Improved Heart Disease Prediction (2024)	Integrate ensemble learning with Bayesian optimization for hyperparameter tuning and SHAP for transparent clinical interpretability.	AdaBoost; Random Forest; XGBoost; Bayesian Optimization; SHAP.	Cleveland (n = 303; 14 attributes); Framingham (n = 5,209; 16 attributes).	Cleveland: Accuracy 0.984, Sensitivity 0.989 (optimized XGBoost). Framingham: Accuracy 0.959, Sensitivity 0.975 (optimized XGBoost).	Requires real-time pipeline integration, extensive validation, and pilot studies in hospital environments before clinical deployment.

6. Research Gap

The comparative review of existing literature exposes several gaps that the proposed research is designed to address. These gaps, taken together, justify the need for a hybrid AI framework that unites complementary learning paradigms for the analysis of tabular medical data.

1. **Predominance of traditional techniques.** Existing heart disease prediction models rely predominantly on traditional machine learning techniques, which limits their capacity to model the complex, non-linear interactions present in clinical data.
2. **Underuse of advanced AI architectures on tabular data.** Advanced AI models are rarely applied to tabular medical data, despite their proven capacity to capture rich feature relationships in other domains.
3. **Scarcity of hybrid AI frameworks for clinical prediction.** Hybrid AI approaches that combine complementary models remain limited in the literature on heart disease prediction, leaving a clear and well-defined gap that this research aims to fill.

7. Proposed Methodology

The proposed work introduces a hybrid AI framework that integrates two complementary models, each designed to capture distinct aspects of the data. The framework is designed so that each component compensates for the weaknesses of the other, with their predictions combined to yield the final classification. The individual components and their integration are described in the subsections that follow.

7.1 Primary Predictive Model

The primary predictive model is selected for its ability to learn effectively from structured, tabular data. It is designed to exploit patterns inherent in clinical records, handle data imperfections common to medical datasets, and maintain strong predictive performance with computational efficiency. Its principal characteristics, which make it well suited to clinical prediction tasks, are as follows:

- **Ensemble-based prediction.** The model constructs an ensemble of base estimators in which each successive estimator is trained to correct the residual errors of its predecessors, progressively refining predictive performance.
- **Sequential error correction.** By sequentially focusing on previously misclassified instances, the model progressively refines its predictive performance across training iterations.
- **Built-in regularization.** Integrated regularization mechanisms penalize model complexity and help prevent overfitting, which is essential when working with limited medical datasets.

- **Native handling of missing values.** The algorithm accommodates missing values directly during training, a valuable property given the incompleteness common to clinical records.
- **Computational efficiency.** The model is designed for efficient training and inference, making it scalable to datasets of varying size.
- **Interpretable feature importance.** Feature contributions can be quantified through analysis techniques, supporting transparency and clinical interpretability.

7.2 Complementary Predictive Model

The complementary AI model is selected for its ability to model complex dependencies among features within structured data. It applies advanced representational mechanisms to capture interactions between clinical variables that simpler models may overlook. Its key properties are summarized below:

- **Advanced feature representation.** Each input feature is transformed into a rich internal representation that enables the model to learn meaningful patterns beyond raw input values.
- **Inter-feature dependency modeling.** The model captures interactions among features, allowing it to learn which combinations of clinical variables are jointly informative for prediction.
- **Mixed input handling.** The architecture natively accommodates both categorical and numerical inputs, reflecting the heterogeneous nature of medical data.
- **Strong performance on structured data.** On complex tabular prediction tasks, this class of model has been shown to outperform many conventional approaches by capturing higher-order feature relationships.

7.3 Proposed Hybrid Model

The core contribution of this research is a hybrid model that fuses the outputs of the two components described above. The framework operates as two parallel branches whose predicted probabilities are combined through a weighted blend to produce a binary outcome indicating the presence or absence of heart disease.

- **Branch A — Primary Predictive Model.** The first branch exploits the strengths of a structured-data-oriented AI model, focusing on modeling relationships within the tabular feature space and handling data imperfections common to clinical records.
- **Branch B — Complementary Predictive Model.** The second branch employs an advanced AI model that captures complex feature interactions and higher-order dependencies within the patient data.

Weighted blend. The final prediction is obtained by linearly combining the probability estimates of the two branches according to the blending rule:

$$P(y = 1) = \alpha \cdot P(y = 1 \mid \text{Model A}) + (1 - \alpha) \cdot P(y = 1 \mid \text{Model B})$$

Here, $\alpha \in [0, 1]$ is a tunable weighting coefficient that governs the relative contribution of each branch. The resulting blended probability is thresholded to yield the binary prediction, where 0 denotes the absence and 1 the presence of heart disease. By calibrating α , the framework can be tuned to achieve an optimal balance between the two learning paradigms for a given dataset.

8. Evaluation Metrics

The performance of the proposed framework and of all baseline models will be assessed using a set of standard classification metrics. Because the relative cost of different error types varies in a clinical context, several complementary metrics are employed to provide a balanced and reliable assessment of model quality.

8.1 Precision

Precision is the proportion of true positive predictions among all positive predictions made by the model. Higher precision values indicate a lower rate of false positive errors. Precision focuses on the accuracy of positive predictions and is particularly important when the cost of a false positive is high.

8.2 Recall (Sensitivity)

Recall, also known as sensitivity, is the proportion of actual positive cases that are correctly identified by the model. Higher recall values indicate a lower rate of false negative errors. Recall captures the model's ability to find all relevant cases and is especially crucial when the cost of missing a positive case is high, as is the case in medical diagnosis, where a missed diagnosis may have severe consequences.

8.3 Accuracy

Accuracy is the proportion of correct results, comprising both true positives and true negatives, among the total number of cases examined. Higher accuracy values indicate that the model is making more correct predictions overall. While accuracy provides an intuitive measure of overall correctness, it may not be reliable for imbalanced datasets in which one class significantly outnumbers the other, and it is therefore interpreted alongside the other metrics.

8.4 F1-Score

The F1-score integrates precision and recall into a single comprehensive measure of model performance, computed as their harmonic mean. Higher F1-score values indicate a better balance between precision and recall. Because it accounts for both false positives and false negatives, the F1-score is well suited to mitigating the impact of class imbalance and is a primary criterion in the evaluation of the proposed framework.

9. Expected Outcomes

Upon completion, this research is expected to deliver the following outcomes:

- **A validated hybrid framework.** A fully implemented hybrid AI framework integrating two complementary models for heart disease prediction, with a calibrated combination strategy that balances their respective contributions.
- **Demonstrated performance gains.** Empirical evidence, obtained through rigorous benchmarking against state-of-the-art individual models, that the hybrid approach improves predictive accuracy and robustness on standard clinical metrics.
- **Improved modeling of clinical patterns.** A demonstrated capacity to capture non-linear feature interactions and long-range dependencies that single-model approaches fail to represent.
- **A foundation for clinical decision support.** A reproducible methodology that can serve as a basis for future translation into practical clinical decision-support tools.